

Date (Local): 11-July-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: KelleyCruise Location: NE PacificStation: Axial BaseCast #/Filename: Dive 907Event: VISIONS 16Bottom Depth (m): 2605 mStart Time (Local): 10:47:14End Time (Local): 16:18:34Sampling Start (Local): 16:35Start Time (GMT): 17:41:14End Time (GMT): 23:18:34Sampling End (Local): 16:47Start Latitude: 45°49.016135'NEnd Latitude: 45°48.988'NCable Rate Out m/min: N/AStart Longitude: 129°45.255538'WEnd Longitude: 129°45.252'WMax Cable Out (m): N/A

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1	Aft - Starboard			OX-001									SA0100	Triggered
2	Niskin			OX-002									SA0101	21:10.31 (UTC)
3				OX-003									SA0102	
4														
5														
6														
7														
8														
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21														
22														
23														
24														

COMMENTS:

Date (Local): 15 July 2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: KellumCruise Location: NE PacificStation: 250m SW 7 LJO1ACast #/Filename: Dire 913Event: VISIONS 16Bottom Depth (m): 2907 mStart Time (Local): 15:25.14End Time (Local): 16-07-2016 13:51.21Sampling Start (Local): 22:00Start Time (GMT): 22:25.14End Time (GMT): 20:51.21Sampling End (Local): 22:30Start Latitude: 44° 30.569468 NEnd Latitude: 44° 30.5695 NCable Rate Out m/min: N/AStart Longitude: 125.24.2500 WEnd Longitude: 125.24.2500 WMax Cable Out (m): N/A

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 mL (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes
1	Fore	Niskin		OX004									97	Sample volume incl rinses
2				OX005									98	
3	Aft	Niskin		OX007									99	
4				OX008									103	
5														
6														
7														
8														
9														
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11														
12														
13														
14														
15														
16														
17														
18														
19														
20														
21														
22														
23														
24														

COMMENTS:

Date (Local): 17-July-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Endurance

Page 1

Primary Investigators: KelleyCruise Location: NE PacificStation: Array SPCast #/Filename: Dive 918Event: VISIONS 16Bottom Depth (m): 574 mStart Time (Local): 16:29.55End Time (Local): 17:44.27Sampling Start (Local): 19:30Start Time (GMT): 23:29.55End Time (GMT): 00:44.27Sampling End (Local): 20:00Start Latitude: 44° 21.5841 NEnd Latitude: 44° 22.4504 NCable Rate Out m/min: N/AStart Longitude: 124.57.8729 WEnd Longitude: 124.57.4004 WMax Cable Out (m): N/A

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes
1	Forward			OX109			2766 ⁴		CH19	DIC66	2		NS	Sample volume incl rinses
2	Niskin						2765 ⁵		CH20	DIC65	3		NS	Time 00:34.35 (UTC)
3	Jason													
4														
5	AFT Niskin			OX110			2767 ⁴		CH22	DIC71	2		SA106 ⁶	sample order
6	Jason								CH17	DIC72	3		SA107 ⁷	
7														
8														
9														
10														
11														
12														
13														
14														
15														
16														
17														
18														
19														
20														
21														
22														
23														
24														

COMMENTS:

* Sampling began ~ 2000 finished ~ 2030

Date (Local): 08-Aug-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: Kelley / KawkaCruise Location: NE PacificCast #/Filename: Dir J-945Event: VISIONS 16Start Time (Local): 18:16End Time (Local): 23:38:43Start Time (GMT): 01:16End Time (GMT): 06:38:43Start Latitude: 44° 31.333' NEnd Latitude: 44° 31.645' NStart Longitude: 125° 22.843' WEnd Longitude: 125° 22.841' WStation: Slope Base DPBottom Depth (m): 2894Sampling Start (Local): 00:01 09-Aug-2016Sampling End (Local): 00:22 09-Aug-2016Cable Rate Out m/min: NAMax Cable Out (m): NA

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1	Aft	2801	22:33	OX-371					CH-01				S 241	
2	"	2801	22:33	OX-372					CH-02				S 242	
3	Forward	2600	22:45	OX-373A					CH-03				S 243	
4	"	2600	22:45	OX-374					CH-04				S 244	
5														
6														
7														
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22														
23														
24														

COMMENTS:

Fired 8 bottles on the bottom

Date (Local): 15-Jul-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: Kelley/Kawka

Cruise Location: NE Pacific

Station: 250 m SW of LJ01A

Cast #/Filename: CTD-01

Event: VISIONS 16

Bottom Depth (m): 2908 m

Start Time (Local): 0100

End Time (Local): 03:08:32

Sampling Start (Local): 02:15:18

Start Time (GMT): 08:03:00

End Time (GMT): 10:08:32

Sampling End (Local): 02:16

Start Latitude: 44° 30.859' N

End Latitude: 44° 30.858' N

Cable Rate Out m/min: 60 m/min

Start Longitude: 125° 23.422' W

End Longitude: 125° 23.427' W

Max Cable Out (m): 2850 m

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 mL (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 mL	Notes Sample volume incl rinses
1	2850		0215											
2														
3														
4														
5														
6														
7														
8	2850		0215											
9														
10														
11														
12														
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14														
15														
16														
17														
18														
19														
20														
21														
22														
23														
24														

Manual Firing of 8 bottles
at 2850 m
Use for HPIES

COMMENTS: chlorophyll spike @ 40-45m O₂ max - ~30-35m 750m O₂ minimum

Time 8 m bottle for WDIRC 5 min - 1 @ 10m 20 m/min - 1st 100 m

Date (Local): 15-July-2016

R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: KelleyCruise Location: NE PacificStation: 250 m SW of LJOIACast #/Filename: CA-CTD-2016-02Event: VISIONS 16Bottom Depth (m): 2908 mStart Time (Local): 3:57:02End Time (Local): 5:48:23Sampling Start (Local): NAStart Time (GMT): 10:57:02End Time (GMT): 12:48:23Sampling End (Local): NAStart Latitude: 44° 30.856' NEnd Latitude: 44° 30.856' NCable Rate Out m/min: 30 m/min < 100
60 m/min > 100Start Longitude: 125° 23.432' WEnd Longitude: 125° 23.434' WMax Cable Out (m): 2875 m

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 mL (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1	2850		TOP	OFF	OF									
2	"		HPIES		WATER									
3	"													
4														
5														
6														
7														
8														
9														
10														
11														
12														
13														
14														
15														
16														
17														
18														
19														
20														
21														
22														
23														
24														

COMMENTS:

5 min soak @ 10m

30 m/min to 100m
60 m/minO₂ 30-40
Chlor 38-48 broad

Date (Local): 07-16-2016

R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: Kelley/KawkaCruise Location: NE PacificStation: 250m E of SB SP MooringCast #/Filename: CTD03Event: VISIONS 16Bottom Depth (m): 2907mStart Time (Local): 10:11 PDTon deck
End Time (Local): 11:08 PDTSampling Start (Local): 11:13Start Time (GMT): 17:11 UTCEnd Time (GMT): 18:06 UTCSampling End (Local): 11:55Start Latitude: 44° 31.726' NEnd Latitude: 44° 31.726'Cable Rate Out m/min: 30m/min to 220mStart Longitude: -125° 23.201' WEnd Longitude: 125° 23.201'Max Cable Out (m): 220m

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1	220	221					2753		CH-18	DIC 049			5462	
B 2	220	221	Leaks											
3	200	200					2754		CH-26	050			5461	
4	200	200												
5	180	180					2755		CH-25	051			5460	
6	180	180												
7	160	160		0x115			2756		CH-24	052			5459	
8	160	160												
9	140	140		0x116			2757		CH-19	053			5458	
B 10	140	140	Leaks											
11	120	120					2758		CH-23	054			5457	
B 12	120	120												
13	100	100					2759		CH-20	055			5463	
14	100	100												
15	80	80					2760		CH-17	056			5464	
B 16	80	80	Leaks											
17	60	60		0x117			2761		CH-22	057			5465	
B 18	60	60	Leaks						CH/CH					
19	40	40		0x118			2762		21/02	058			5466	
B 20	40	40	Leaks						CH/CH					
21	20	21					2763		03/04	059			5467	
B 22	20	21	Leaks											
23	10	12		119/120			2764		CH 05	060			5468	
B 24	10	10	Leaks											

COMMENTS:

10 min soak @ 10m
 10 min 20m
 35 min 90m
 very low chlorophyll with brom max SWN

Date (Local): 18-July-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: Kelley / KawkaCruise Location: NE PacificStation: 250 m E of SP at Slope BaseCast #/Filename: CTD-04Event: VISIONS 16Bottom Depth (m): 2907 mStart Time (Local): 11:45:23End Time (Local): 12:17Sampling Start (Local): N/AStart Time (GMT): 18:45:23End Time (GMT): 19:17Sampling End (Local): N/AStart Latitude: 44° 31.736'End Latitude: 44° 31.729' NCable Rate Out m/min: 30 m/minStart Longitude: 125° 23.107'End Longitude: 125° 23.084' WMax Cable Out (m): 220 m

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1	220													
B 2 ✓	220													
3	200													
4	200													
5	180													
6	180													
7	160													
8	160													
9	140													
B 10 ✓	140													
11	120													
B 12	120													
13	100													
14	100													
15	80													
B 16 ✓	80													
17	60													
B 18 ✓	60													
19	40													
B 20 ✓	40													
21	20													
B 22 ✓	20													
23	10													
B 24 ✓	10													

COMMENTS:

10 min hold @ 10m ↓ 30m/min to 220m + 30m/min ↑

Date (Local): 18-July 2016

R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: Kelley / KowkaCruise Location: NE PacificStation: 250 m E of SP @ Slope BaseCast #/Filename: CTD-05Event: VISIONS 16Bottom Depth (m): 2907Start Time (Local): 12:22End Time (Local): 12:41Sampling Start (Local): N/AStart Time (GMT): 19:22End Time (GMT): 19:41Sampling End (Local): N/AStart Latitude: 44° 31.729' NEnd Latitude: 44° 31.729' NCable Rate Out m/min: 30m/minStart Longitude: 125° 23.084' WEnd Longitude: 125° 23.084' WMax Cable Out (m): 220 m

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1	220													
2	220													
3	200													
4	200													
5	180													
6	180													
7	160													
8	160													
9	140													
10	140													
11	120													
12	120													
13	100													
14	100													
15	80													
16	80													
17	60								2					
18	60													
19	40			2					2					
20	40													
21	20			2										
22	20													
23	10													
24	10													

COMMENTS:

30m/min from 10m

chlorophyll max 40-60

O₂ - max 40 ✓

Date (Local): 18-July-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: Kelly / KawkaCruise Location: NE PacificStation: 250 m E of SP@ Slope BaseCast #/Filename: CTD-06Event: VISIONS 16on deck Bottom Depth (m): 2907Start Time (Local): 12:44End Time (Local): 13:18Sampling Start (Local): 1337Start Time (GMT): 19:44End Time (GMT): 20:18Sampling End (Local): 1441Start Latitude: 44° 31.729' NEnd Latitude: 44° 31.729' NCable Rate Out m/min: 30 m/minStart Longitude: 125° 23.084' WEnd Longitude: 125 23.084' WMax Cable Out (m): 220 m

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 mL (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes
1	220	✓					'2768		CH-1	'045			'0097	OVER SAMPLED FROM PREVIOUS CAST
2	220	✓	Leaking											
3	200	✓					'2776		CH-2	'041			'0109	
4	200	✓												
5	180	✓					'2775		CH-3	'037			'0110	
6	180	✓												
7	160	✓		'0x111			'2774		CH-4	'033			'0112	
8	160	✓												
9	140	✓		'0x112			'2773		CH-5	'029			'0111	
10	140	✓	Leaking											
11	120	✓					'2772		CH-6	'025			'0113	Reset after
12	120	✓	Leaking											#11 to 6a data
13	100	✓					'2771		CH-7	'026			'0114	
14	100	✓												
15	80	✓					'2770		CH-8	'030			'0115	
16	80	✓	Leaking											
17	60	✓		'0x113			'2769		CH-18 CH-21	'034			'0116	
18	60	✓	Leaking											
19	40	✓		20x114 0x103A			'2777		CH-23 CH-24	'038			'0117	
20	40	✓	Leaking											
21	20	✓		'0x105			'2778		CH-25	'042			'0118	
22	20	✓	Leaking											
23	10	✓		20x106 0x107			'2779		CH-26	'046			'0119	
24	10	✓	Leaking											

COMMENTS:

Date (Local): 19-July 2016

R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: Kelley/KawkaCruise Location: NE PacificStation: 250 m E of EA Shallow ProfilerCast #/Filename: CTD - 07Event: VISIONS 16Bottom Depth (m): 574mStart Time (Local): 14:46End Time (Local): 15:16Sampling Start (Local): N/AStart Time (GMT): 21:46End Time (GMT): 22:16Sampling End (Local): N/AStart Latitude: 44° 22.464' NEnd Latitude: 44° 22.464' NCable Rate Out m/min: 30 m/minStart Longitude: 124° 57.204 WEnd Longitude: 124.57.204 WMax Cable Out (m): 565m

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1														
2														
3														
4														
5														
6														
7														
8														
9														
10														
11														
12														
13														
14														
15														
16														
17														
18														
19														
20														
21														
22														
23														
24														

No Samples Taken

COMMENTS:

To 220m

Date (Local): 19-July 2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: Kelley/KawkaCruise Location: NE PacificStation: 250 m F of EA Shallow ProfileCast #/Filename: CTD - 08Event: VISIONS16Bottom Depth (m): 574mStart Time (Local): 15:18End Time (Local): 15:54Sampling Start (Local): N/AStart Time (GMT): 22:18End Time (GMT): 22:54Sampling End (Local): N/AStart Latitude: 44° 22.464' NEnd Latitude: 44° 22.464' NCable Rate Out m/min: 30m/minStart Longitude: 124° 57.204' WEnd Longitude: 124° 57.203' WMax Cable Out (m): 565m

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1														
2														
3														
4														
5														
6														
7														
8														
9														
10														
11														
12														
13														
14														
15														
16														
17														
18														
19														
20														
21														
22														
23														
24														

No Samples Collected

COMMENTS:

To 550m

Chlorophyll max ~ 40m

Up @ 60m/min to 100m then 30m/min to 10m

Date (Local): 19-July 2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: Kelley / KawkaCruise Location: NE PacificStation: 250 m E of EA Shallow ProfilerCast #/Filename: CTD - 09Event: VISIONS16Bottom Depth (m): 574 mStart Time (Local): 15:56End Time (Local): 16:50Sampling Start (Local): 1704Start Time (GMT): 22:56End Time (GMT): 00:56Sampling End (Local): 1800Start Latitude: 44° 22' 46.5" NEnd Latitude: 44° 22' 46.5" NCable Rate Out m/min: 30 m/minStart Longitude: 124° 57.204' WEnd Longitude: 124° 57.204' WMax Cable Out (m): 560 m

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1	550			108						039'			573/74	
2	550													
3	400			102						040'			575'	
4	400													
5	200			101A			2789	CH-1	043'				576'	
6	200													
7	180						2780	CH-2	027'				577'	
8	180													
9	40						2781	CH-3	028'				578'	
10	14													
11	120						2782	CH-4	031'					
12	120													
13	100						2783	CH-5	032'					
14	100													
15	80						2784	CH-6	035'					
16	80													
17	60						2785	CH-7	036'					
18	60													
19	40			100B 97			2786	CH-19	044'				579/0081	
20	40			2				CH-20						
21	20						2787	CH-21	047'					
22	20													
23	10			098A 97			2788	CH-23	048'				0080/0082	
24	10			2										

COMMENTS:

To 550 m

* Nutrients taken on
19, 21, 23 before DIC

Date (Local): 28-July-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

Page 1

Primary Investigators: D. Kelley / O. KawkaCruise Location: NE PacificStation: 250 m NW of Shallow Profiler at Coral RidgeCast #/Filename: CTD-10Event: VISIONS 16Bottom Depth (m): 2618 mStart Time (Local): 10:33End Time (Local): 11:04Sampling Start (Local): No Samples TakenStart Time (GMT): 17:33End Time (GMT): 18:04Sampling End (Local): 11 11 11Start Latitude: 45° 49.914' NEnd Latitude: 45° 49.914' NCable Rate Out m/min: 30 m/min - down + upStart Longitude: 129° 45.329' WEnd Longitude: 129.45.328' WMax Cable Out (m): 220 m (Seabird)

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 mL (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 mL	Notes Sample volume incl rinses
1														
2														
3														
4														
5														
6														
7														
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10														
11														
12														
13														
14														
15														
16														
17														
18														
19														
20														
21														
22														
23														
24														

COMMENTS:

soak at 10m for 10 min

final depth 220 m

O₂ - 20m + 60m - 20mchlorophyll max 65m
range 65-85

Date (Local): 28-July-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

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Primary Investigators: Kelley / KawkaCruise Location: NE PacificStation: 250 m NW of SP @ Axial BaseCast #/Filename: CTD-11Event: VISIONS 16Bottom Depth (m): 2618 mStart Time (Local): 11:06End Time (Local): 11:23Sampling Start (Local): No Samples TakenStart Time (GMT): 18:06End Time (GMT): 18:23Sampling End (Local): " " "Start Latitude: 45° 49.915' NEnd Latitude: 45° 49.914' NCable Rate Out m/min: 30 m/min - up + downStart Longitude: 129° 45.329' WEnd Longitude: 129° 45.328' WMax Cable Out (m): 220m (by Seabird)

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1														
2														
3														
4														
5														
6														
7														
8														
9														
10														
11														
12														
13														
14														
15														
16														
17														
18														
19														
20														
21														
22														
23														
24														

COMMENTS:

Chlor max 60m Range 50-90

O₂ 20m + 60m

Date (Local): 28-July-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

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Primary Investigators: Kelley / KawkaCruise Location: NE PacificStation: 250 m NW of SP at Axel BaseCast #/Filename: CTD-12Event: VISIONS 16Bottom Depth (m): 2618Start Time (Local): 11:25End Time (Local): 11:52Sampling Start (Local): 1225Start Time (GMT): 18:25End Time (GMT): 18:52Sampling End (Local): 1330Start Latitude: 45° 49.914' NEnd Latitude: 45° 49.914' NCable Rate Out m/min: 30 m/minStart Longitude: 125° 45.330' WEnd Longitude: 129° 49.329' WMax Cable Out (m): 220 m (by Seabird)

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 mL (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1	220	✓					'2790		CH-26	'17			'89	
2	220	✓												
3	200	✓					'2791		CH-8	'16			'88	
4	200	✓												
5	180	✓					'2792		CH-3	'15			'87	
6	180	✓												
7	160	✓					'2793		CH-20	'14			'86	
8	160	✓												
9	140	✓					'2794		CH-5	'80			'85	
10	140	✓												
11	120	✓					'2795		CH-23	'90			'91	
12	120	✓												
13	100	✓					'2796		CH-2	'81			'92	
14	100	✓												
15	80	✓					'2798		CH-4	'88			'93	
16	80	✓												
17	60	✓					'2797		CH-7	'89			'94/95	
18	60	✓												
19	40	✓		21361 362			'2799		CH-1	'82			'96/473	
20	40	✓												
21	20	✓		21363 364			'2800		CH-25	'85			'472	
22	20	✓												
23	10	✓		21365 366			'2801		CH-21	'92			'474	
24	10	✓												

COMMENTS:

Chlor max 60m
Range
45-85

Date (Local): 28-July-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

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Primary Investigators: Kelley / KawkaCruise Location: NE PacificStation: 500 m N DP at Axial BaseCast #/Filename: CTD-13Event: VISIONS 16Bottom Depth (m): 2605Start Time (Local): 13:44End Time (Local): 15:39Sampling Start (Local): no maskingStart Time (GMT): 20:44End Time (GMT): 22:39Sampling End (Local): sampledStart Latitude: 45° 50.044' NEnd Latitude: 45° 50.044' NCable Rate Out m/min: 30 m/min 1st 100 then 60 m/minStart Longitude: 129° 45.542' WEnd Longitude: 129.45.542' W~~Max Cable Out (m):~~ 2594 m (Seabird)

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1	2594													
2	2594													
3	2594													
4	2044													
5														
6														
7														
8														
9														
10														
11														
12														
13														
14														
15														
16														
17														
18														
19														
20														
21														
22														
23														
24	2044													

COMMENTS: chlor max ~ 60 m 45-80 30 m/min 1st 100 m
 10 min soak @ 10 m, up to just below surface 60 m/min after that to 2550 m

Date (Local): 6-Aug-2016 R/V Sikuliaq CA-OOI Cruise SKQW201610S - CTD Data Sheet

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Primary Investigators: Kelley / KawkaCruise Location: NE PacificStation: 250 m E of DPOIB-2016Cast #/Filename: CTD-14Event: VISIONS 16Bottom Depth (m): 578 mStart Time (Local): 23:51 06-Aug-2016End Time (Local): 01:10 07-Aug-2016Sampling Start (Local): 01:15Start Time (GMT): 06:51 (07-Aug-2016)End Time (GMT): 08:10 07-Aug-2016Sampling End (Local): 01:43Start Latitude: 44° 22.110' NEnd Latitude: 44° 22.10' NCable Rate Out m/min: 30 m/minStart Longitude: 124° 56.959' WEnd Longitude: 124° 56.959' WMax Cable Out (m): 572.2 m

Niskin Bottle No.	Target Depth (m)	Actual Depth (m)	Time (Local)	Oxygen 600 mL	Other Gases 1	Other Gases 2	Nutrients 70 mL (2/3 full)	Biogenic Si 70 ml (2/3 full)	Chloroph 500 mL	DIC 12 oz Bottle	Phyto counts 300 mL	Bacteria	Salinity 500 ml	Notes Sample volume incl rinses
1	570 ✓												5475	
2	570 ✓													
3	540 ✓													
4	540 ✓								CH01					
5	400 ✓								CH02				5476	
6	400 ✓													
7	300 ✓													
8	300 ✓													
9	250 ✓			20x367 0x368										
10	250 ✓													
11	200 ✓								CH-03				5477	
12	200 ✓													
13	110 ✓			0x369 20x370										
14	110 ✓													
15	50 ✓												5478	
16	50 ✓													
17	20 ✓								CH04					
18	20 ✓													
19	10 ✓								CH05				5479	
20	10 ✓													
21														
22														
23														
24														

COMMENTS:

5 min soak @ 7 m up to just below surface 1st 100 30 m/min
 then 60 m/min til 270 m